

Using The LIBOR Market Model to Price The Interest Rate Derivatives: A Recombining Binomial Tree Methodology

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AGENDA

- ✗ Introduction
- ✗ Review of interest rate models
- ✗ Introducing the HSS recombining node methodology and applying to the LIBOR market model
- ✗ The pricing of the interest rate derivatives in LMM
- ✗ Conclusions

INTRODUCTION

- ✗ We can't observe the instantaneous short rate and instantaneous forward rate in the traditional interest rate models
- ✗ We adapt LIBOR market model which is based on the forward LIBOR rate that we can observe from the daily market
- ✗ When implementing LMM in lattice method, we face the explosive tree due to the non-Markov property

INTRODUCTION

- ✘ The nonrecombining nodes make our pricing procedure inefficient and don't satisfy the market requirement
- ✘ We apply the HSS methodology into LMM that make the nodes combine and make the pricing the derivatives feasible
- ✘ The method we proposed make our valuation more efficient and more accurate

REVIEW OF INTEREST RATE MODELS

- Equilibrium models
- No-arbitrage models
 - × Instantaneous short rate models
 - × Instantaneous forward rate model
 - × Forward rate model

FORWARD RATE MODEL

LIBOR MARKET MODEL (LMM)

- ✗ The LMM was discovered by Brace, Gatarek, and Musiela (1997) and was initially referred to as the BGM model by practitioners
- ✗ There are two commonly used versions of the LMM:
 - + one is the lognormal forward LIBOR model (LFM) for pricing caps
 - + the other is the lognormal swap model (LSM) for pricing swaptions

FORWARD RATE MODEL

LIBOR MARKET MODEL (LMM)

- ✗ The LFM specifies the forward rate $f(t; T_i, T_{i+1})$ following zero-drift stochastic process under its own forward measure:

$$\frac{df(t; T_i, T_{i+1})}{f(t; T_i, T_{i+1})} = \sigma_i(t) dW_i(t)$$

- + $dW_i(t)$ is a Brownian motion under the forward measure $\tilde{P}^i$ defined with respect to the numeraire asset
- + $\sigma_i(t)$ measures the volatility of the forward rate process

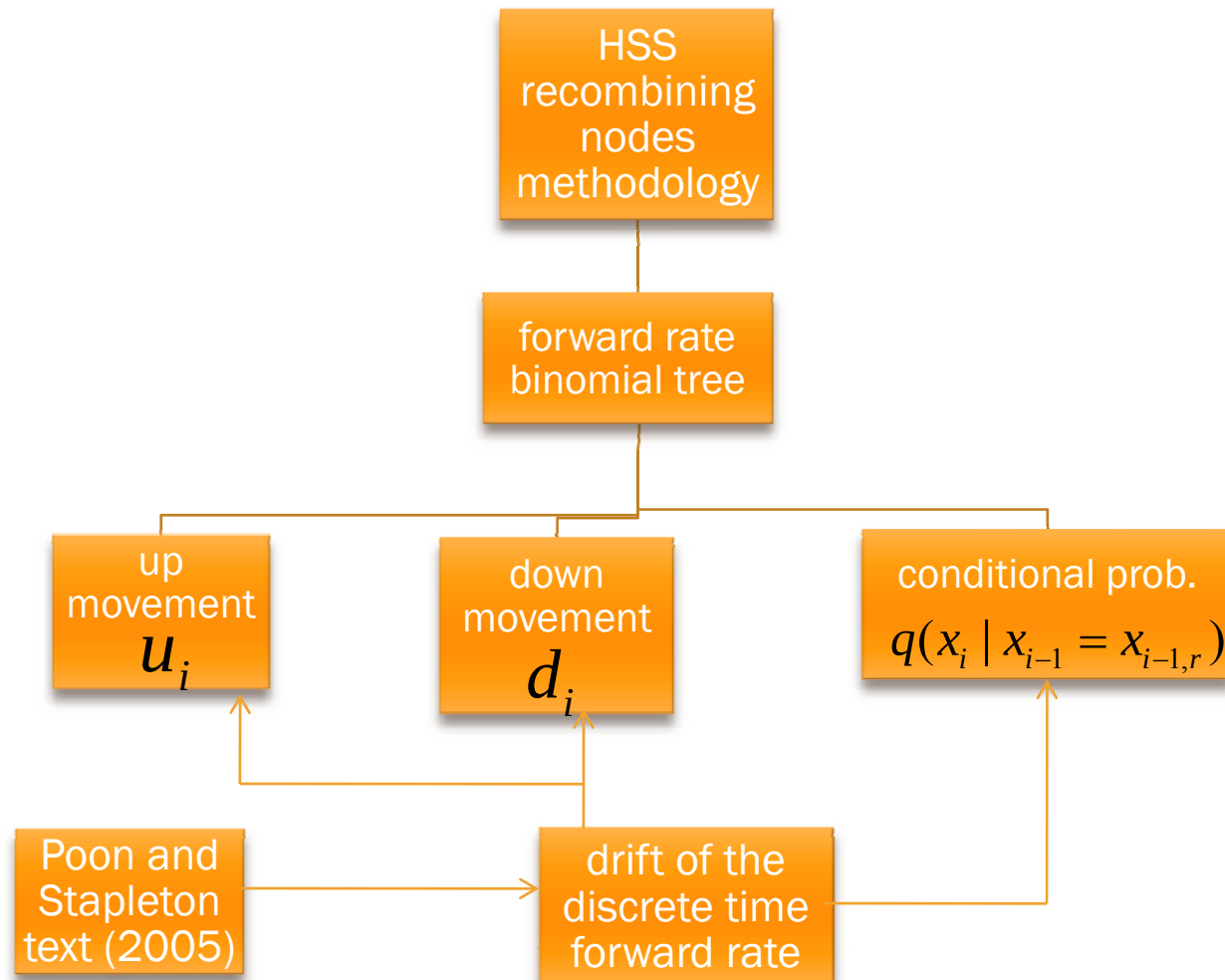
MARKET CONVENTIONS OF THE LMM

- ✗ The relationship between the discrete LIBOR rate $L(T_i, T_{i+1})$ for the term $\delta_i = T_{i+1} - T_i$ and the zero coupon bond price $P(T_i, T_{i+1})$, given as follows:

$$P(T_i, T_{i+1}) = \frac{1}{1 + \delta_i L(T_i, T_{i+1})}$$

where $t = T_0 < T_1 < T_2 < \dots < T_n$ is the time line and δ_i is called the tenor for the period T_i to T_{i+1}

METHODOLOGY



INTRODUCING THE HSS RECOMBINING NODE METHODOLOGY

- ✖ Ho, Stapleton and Subrahmanyam (1995) [HSS] suggest a general methodology for creating a recombining multi-variate binomial tree to approximate a multi-variate lognormal process
- ✖ Our assumption about LMM satisfies the conditions in the HSS methodology and apply it into LMM

HSS METHODOLOGY

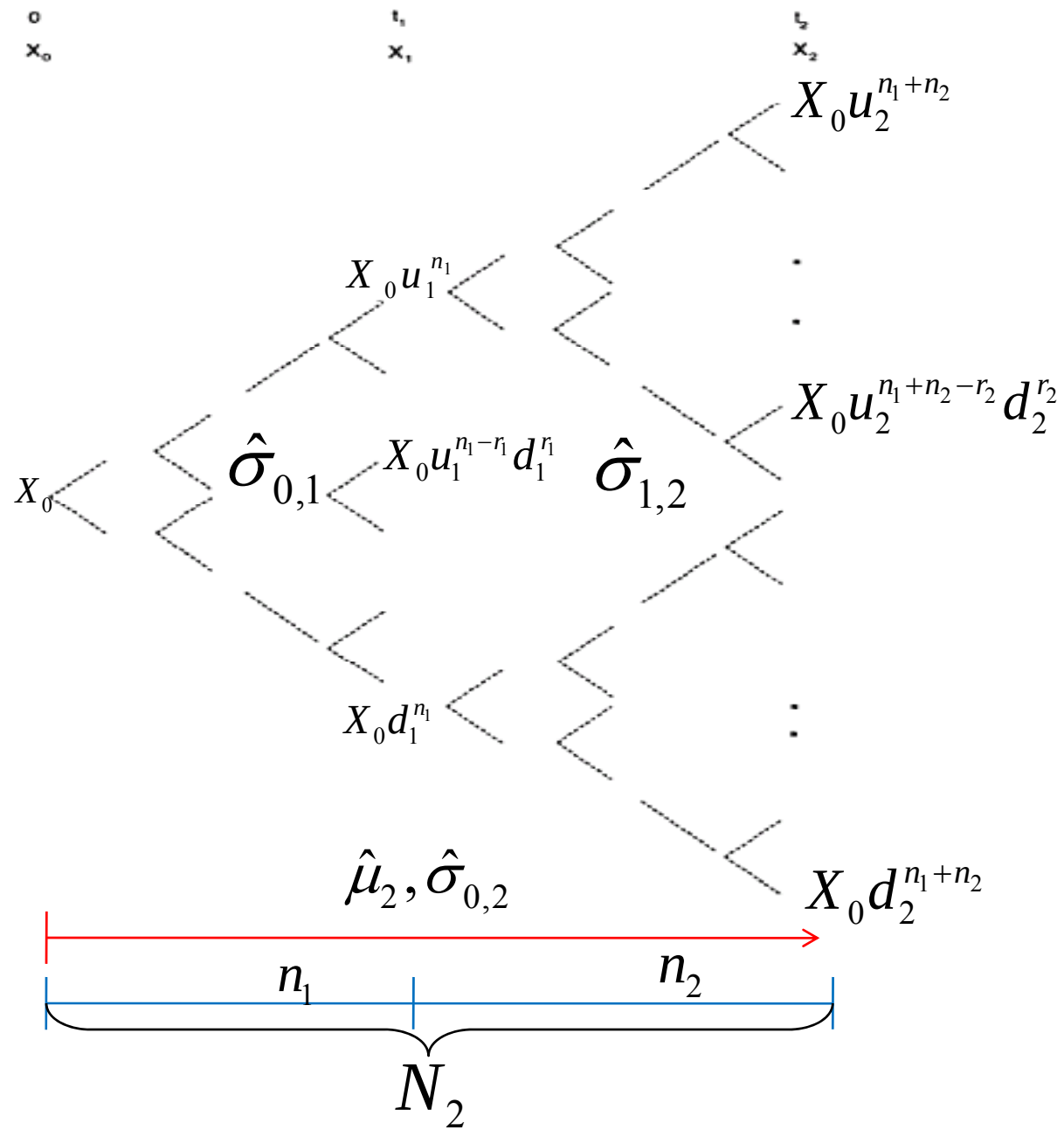
- ✗ HSS assume the price of underlying asset follows a lognormal diffusion process:

$$d \ln X(t) = \mu(X(t), t)dt + \sigma(t)dW(t)$$

where μ and σ are the instantaneous drift and volatility of $\ln X$, and $dW(t)$ is a standard Brownian motion

- + They denote the unconditional mean of the logarithmic asset return at time i as μ_i
- + The conditional volatility over the period $i-1$ to i is denoted as $\sigma_{i-1,i}$
- + The unconditional volatility is $\sigma_{0,i}$

TIME



HSS METHODOLOGY

- ✖ HSS method involves the construction of m separate binomial distribution, where the time periods are denoted $t_1, \dots, t_i, \dots, t_m$
- ✖ And have the set of a discrete stochastic for X_i , where X_i is only defined at time t_i
- ✖ The general form of X_i at node r

$$X_{i,r} = X_0 u_i^{N_i - r} d_i^r$$

where $N_i = \sum_{l=1}^i n_l$

HSS METHODOLOGY

- ✗ They denote $x_i = \ln(X_i / X_0)$ and the probabilities to reach x_i given a node $x_{i-1,r}$ at t_{i-1} as

$$q(x_i \mid x_{i-1} = x_{i-1,r}) \text{ or } q(x_i)$$

- ✗ **Lemma 1** Suppose that the up and down movements u_i and d_i are chosen so that

$$d_i = \frac{2(E(X_i) / X_0)^{\frac{1}{N_i}}}{1 + \exp(2\sigma_{i-1,i} \sqrt{(t_i - t_{i-1}) / n_i})}, \quad i = 1, 2, \dots, m,$$

$$u_i = 2(E(X_i) / X_0)^{\frac{1}{N_i}} - d_i, \quad i = 1, 2, \dots, m,$$

HSS METHODOLOGY

Theorem

Suppose that the X_i are joint lognormally distributed. If the X_i are approximated with binomial distributions with $N_i = N_{i-1} + n_i$ stages and u_i and d_i given by the above, and if the conditional probability of an up movement at node r at time $i-1$ is

$$q(x_i | x_{i-1} = x_{i-1,r}) = \frac{a_i + b_i x_{i-1,r} - (N_{i-1} - r) \ln u_i - r \ln d_i}{n_i (\ln u_i - \ln d_i)} - \frac{\ln d_i}{\ln u_i - \ln d_i}, \quad \forall i, r$$

then $\hat{\mu}_i \rightarrow \mu_i$ and $\hat{\sigma}_{0,i} \rightarrow \sigma_{0,i}$ and $\hat{\sigma}_{i-1,i} \rightarrow \sigma_{i-1,i}$ as $n_i \rightarrow \infty$, $\forall i$

APPLYING HSS METHODOLOGY INTO LMM

- ✘ We apply this methodology into the LMM and make some change to satisfy our conventions
- ✘ We have the following propositions

PROPOSITION 1

- ✗ For the forward LIBOR rate which follows the lognormal distribution, we can choose the proper up and down movements to determine the i -th period of the T_n -maturity forward LIBOR rate and have the form

$$f(i; T_n, T_{n+1})_r = f(0; T_n, T_{n+1}) u_i^{N_i - r} d_i^r, \quad i = T_1, T_2, \dots, T_n$$

where
$$d_i = \frac{2[E(f(i; T_n, T_{n+1})) / f(0; T_n, T_{n+1})]^{1/N_i}}{1 + \exp(2\sigma_{i-1,i} \sqrt{(T_i - T_{i-1}) / n_i})}$$

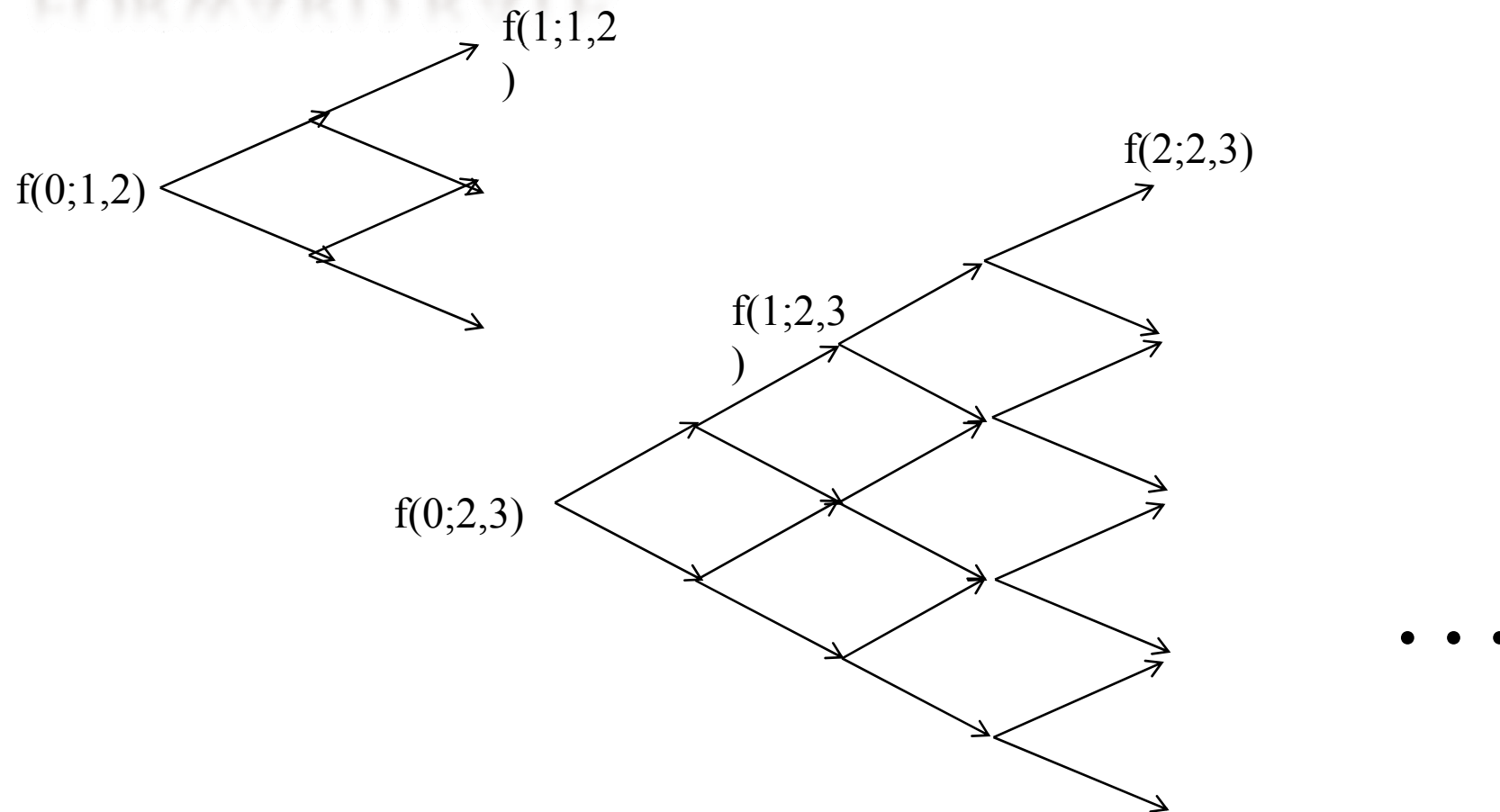
$$u_i = 2[E(f(i; T_n, T_{n+1})) / f(0; T_n, T_{n+1})] - d_i$$

$$N_i = N_{i-1} + n_i$$

r : node's number from top to bottom at time T_i

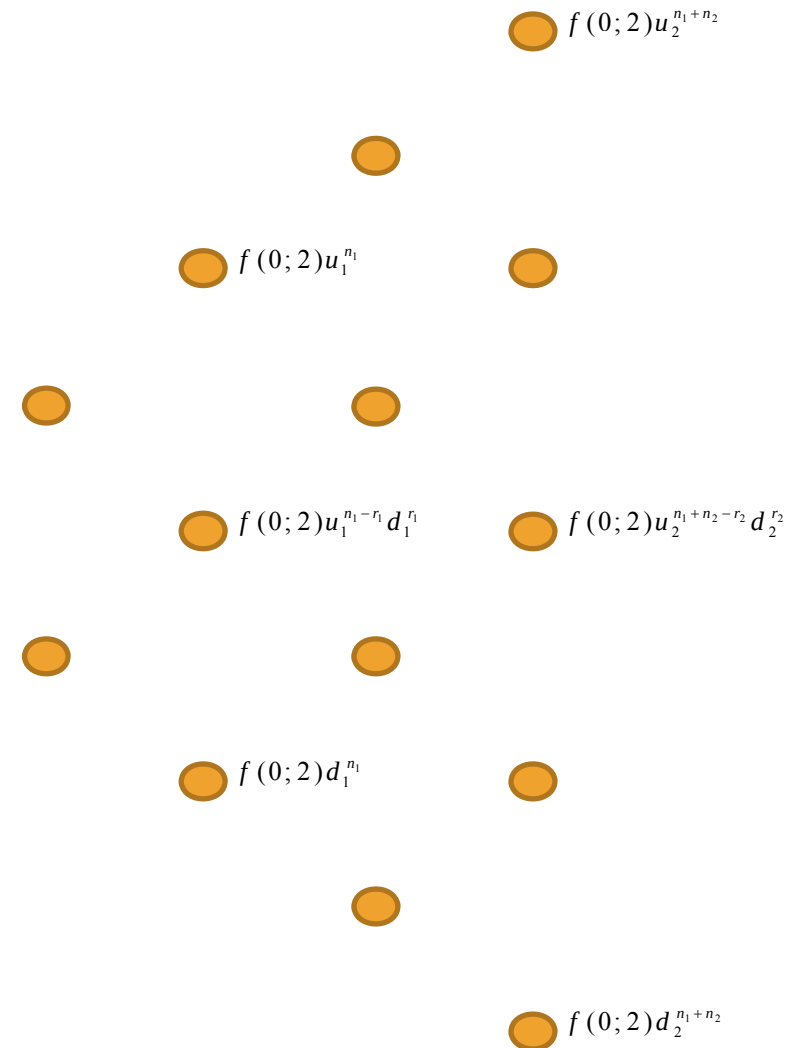
THE BINOMIAL TREES IN DISCRETE TIME

FORWARD RATE



THE STRUCTURE OF THE BINOMIAL TREE

- ✗ With $n_1 + 1$ nodes at T_1
numbered $r = 0, 1, \dots, n_1$.
There are $n_1 + n_2 + 1$ nodes
at T_2 numbered $r = 0, 1, \dots, n_1 + n_2$
Here we write the
forward rate $f(0; 2, 3)$
in abbreviated form $f(0; 2)$
and take $n_1 = n_2 = 2$,
 $0 \leq r_1 \leq n_1, 0 \leq r_2 \leq n_1 + n_2$



PROPOSITION 2

✕ Suppose that the forward LIBOR rate $f(i; T_n, T_{n+1})$ are joint lognormally distributed. If the

$f(i; T_n, T_{n+1})$, $i = T_1, T_2, \dots, T_n$ are approximated with binomial distributions with $N_i = N_{i-1} + n_i$ stages and u_i and d_i given by proposition 1, and if the conditional probability of an up movement at node r at time T_i is

$$q(x_i \mid x_{i-1} = x_{i-1,r}) = \frac{E_{i-1}(x_i) - (N_{i-1} - r) \ln u_i - r \ln d_i}{n_i (\ln u_i - \ln d_i)} - \frac{\ln d_i}{\ln u_i - \ln d_i} \quad \forall i, r$$

PROPOSITION 2

where

$$x_i = \ln \frac{f(i; T_n, T_{n+1})}{f(0; T_n, T_{n+1})}$$

$$E_{i-1}(x_i) = E(x_i) - b_i E(x_{i-1}) + b_i x_{i-1,r}$$

$$b_i = \sqrt{[t_i \sigma_{0,i}^2 - (t_i - t_{i-1}) \sigma_{i-1,i}^2] / t_{i-1} \sigma_{0,i-1}^2}$$

- ✖ To determining the conditional probability, it has some skills to use for the term $E_{i-1}(x_i)$
- ✖ We first derive term $E(x_i)$. Since forward rate is lognormally distributed, we have

$$E(x_i) = \ln \left[\frac{E(f(i; T_n, T_{n+1}))}{f(0; T_n, T_{n+1})} \right] - \frac{1}{2} \sigma_{0,i}^2$$

THE DERIVATION OF EXPECTATION

- ✗ The most important result in Poon and Stapleton text is the drift of the discrete time forward rate and we rewrite as

$$\frac{E_t[f(T_1; T_n, T_{n+1})]}{f(t; T_n, T_{n+1})} = 1 + \frac{\delta_1 f(t; T_1, T_2)}{1 + \delta_1 f(t; T_1, T_2)} \cdot \tilde{\sigma}_{1,n} + \frac{\delta_2 f(t; T_2, T_3)}{1 + \delta_2 f(t; T_2, T_3)} \cdot \tilde{\sigma}_{2,n} + \dots + \frac{\delta_n f(t; T_n, T_{n+1})}{1 + \delta_n f(t; T_n, T_{n+1})} \cdot \tilde{\sigma}_{n,n}$$

- ✗ Then multiple the $\frac{f(t; T_n, T_{n+1})}{f(0; T_n, T_{n+1})}$ term on both side to get the general form of $E_t(f(T_1; T_n, T_{n+1})) / f(0; T_n, T_{n+1})$

THE DERIVATION OF EXPECTATION

✖ The general form of expectation is given as

$$\frac{f(t; T_n, T_{n+1})}{f(0; T_n, T_{n+1})} \times \frac{E_t[f(T_1; T_n, T_{n+1})]}{f(t; T_n, T_{n+1})} =$$
$$\frac{f(t; T_n, T_{n+1})}{f(0; T_n, T_{n+1})} \times \left(1 + \frac{\delta_1 f(t; T_1, T_2)}{1 + \delta_1 f(t; T_1, T_2)} \cdot \tilde{\sigma}_{1,n} + \frac{\delta_2 f(t; T_2, T_3)}{1 + \delta_2 f(t; T_2, T_3)} \cdot \tilde{\sigma}_{2,n} + \dots + \frac{\delta_n f(t; T_n, T_{n+1})}{1 + \delta_n f(t; T_n, T_{n+1})} \cdot \tilde{\sigma}_{n,n} \right)$$

SPECIAL CASE

- ✗ When n_l stages approach infinite $l = 1, \dots, i$, the sum of n_l stages also approach infinite, that is

$$N_i = \sum_{l=1}^i n_l \rightarrow \infty$$

- ✗ We can reduce the up and down movements to the briefer form which is easier to calculate and show as follows

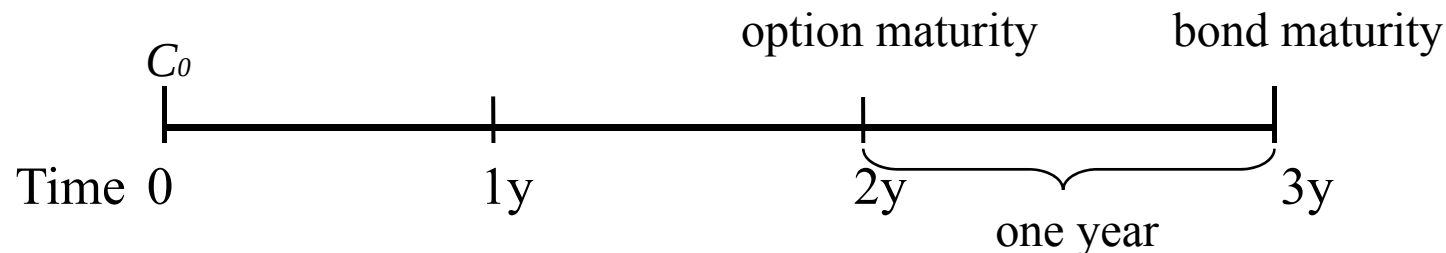
$$d_i = \frac{2}{1 + \exp(2\sigma_{i-1,i} \sqrt{(T_i - T_{i-1}) / n_i})}$$

$$u_i = 2 - d_i$$

- ✗ The conditional probability $q(x_l) \rightarrow 0.5$ as $n_l \rightarrow \infty$

THE VALUATION OF EMBEDDED OPTION ON ZERO COUPON BOND IN LMM

- ✖ The embedded option on ZCB is a bond that can be callable before maturity date with a callable price K



- ✖ We have a three years maturity zero coupon bond with a callable price K equal to 0.952381 dollar at year two

THE VALUATION OF EMBEDDED OPTION ON ZERO COUPON BOND IN LMM

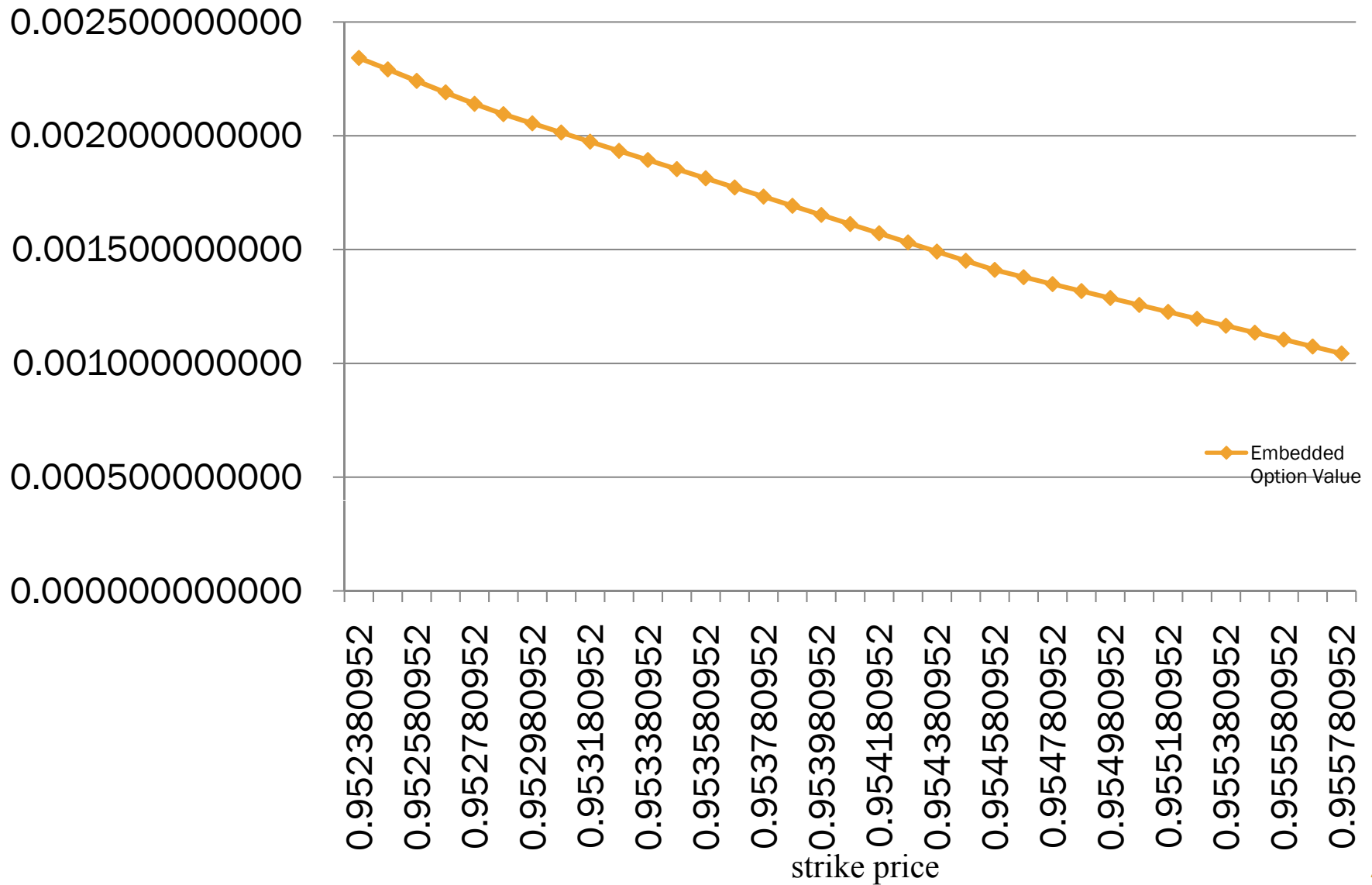
- ✖ Pricing the option value C_0 of this callable bond at time 0
- ✖ Here, we take the flat forward rate 5% and constant volatility 10%

$$\begin{aligned}C_0 &= P(0, 2) \times E[\max(P(2, 3) - K, 0)] \\&= 0.90702948 \times 0.00258128 \\&= 0.00234130\end{aligned}$$

SENSITIVE ANALYSIS – STRIKE PRICE

- ✘ We change strike price from 0.952381 to 0.96 which vary 0.001 to see the relationship between strike price and embedded option value
- ✘ We find that when the strike price increases, the embedded option value decreases

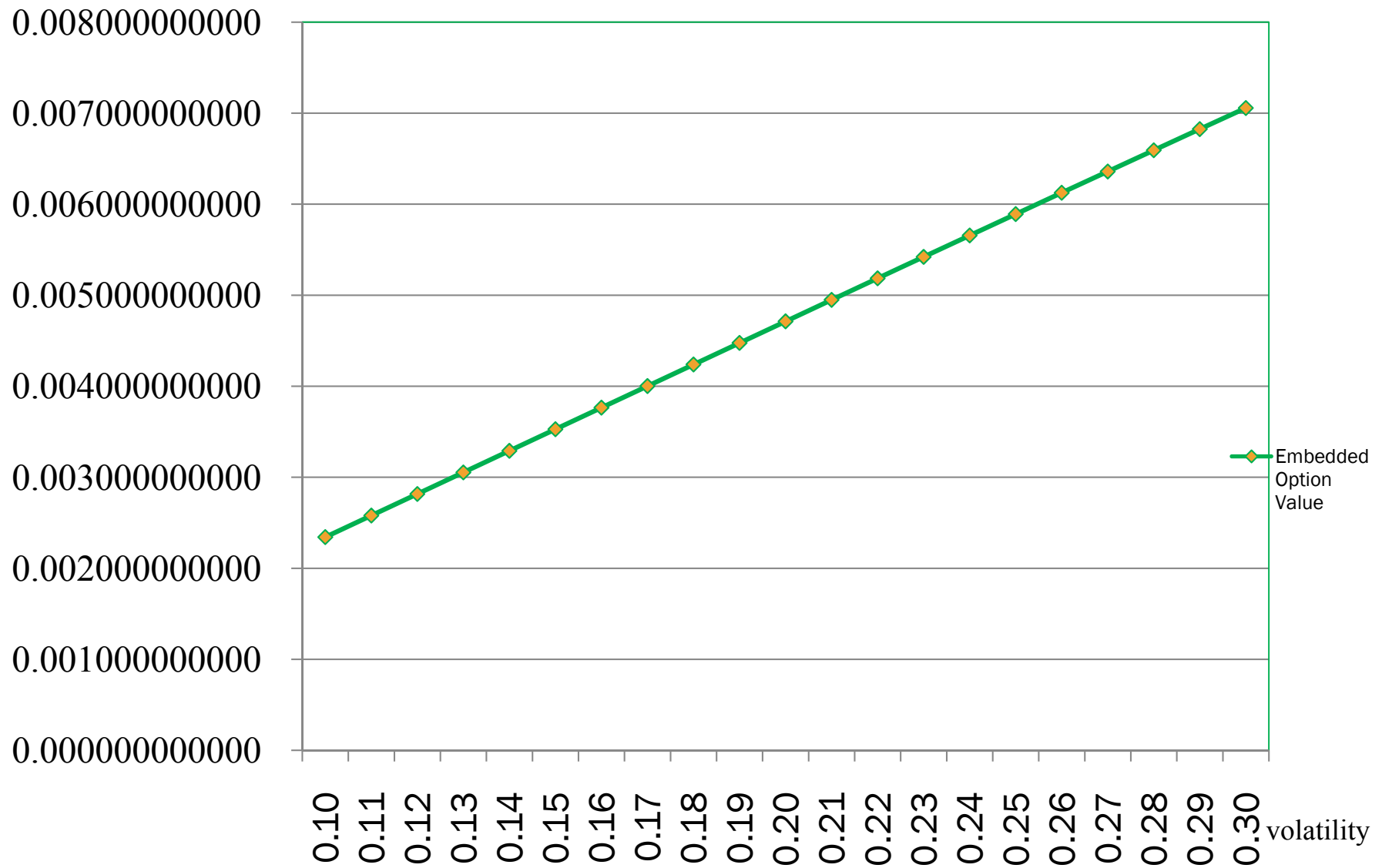
Embedded Option Value



SENSITIVE ANALYSIS – VOLATILITY

- ✖ We increase 1% of the volatility from 10% until reach 30% to see the impact of volatility on the embedded option value
- ✖ We find that when the volatility increases, the embedded option value increases
- ✖ It is consistent with the inference for the Greek letter *vega* when the underlying asset's volatility increases the option value increases, too.

Embedded Option Value



THE VALUATION OF CAPLETS IN LMM

- ✖ The payoff function of a caplet at time T_{i+1} is

$$A \times \delta \times \max(f(T_i; T_i, T_{i+1}) - K, 0)$$

- ✖ It is a caplet on the spot rate observed at time T_i with payoff occurring at time T_{i+1}
- ✖ The cap is a portfolio consisted of n such call options which the underlying is known as caplet

THE THEORETICAL VALUE OF CAPLET

- ✖ We use the Black's formula for the caplet to get the theoretical value of caplet and rewrite as

$$caplet_i(t) = A \times \delta_i \times P(t, T_{i+1}) [f(t; T_i, T_{i+1}) N(d_1) - KN(d_2)]$$

where

$$d_1 = \frac{\ln(f(t; T_i, T_{i+1}) / K) + \sigma_i^2 (T_i - t) / 2}{\sigma_i \sqrt{T_i - t}},$$

$$d_2 = \frac{\ln(f(t; T_i, T_{i+1}) / K) - \sigma_i^2 (T_i - t) / 2}{\sigma_i \sqrt{T_i - t}},$$

NUMERICAL METHOD OF CAPLET

- ✖ We use the payoff function to compute the price in lattice method
- ✖ To get the payoff function at time T_{i+1} , we have to know the evolution of the forward rate $f(0; T_i, T_{i+1})$ at time T_i
- ✖ We construct the binomial tree of $f(0; T_i, T_{i+1})$ and known the $(f(T_i; T_i, T_{i+1})_r - K)^+$, $r = 0, 1, \dots, i$.
- ✖ Calculating the expectation of the payoff at time T_{i+1} and then multiple the ZCB of $P(t, T_{i+1})$ to get the caplet value at time t .

EXAMPLE

- ✖ Assume $\delta = A = 1$, $\sigma = 10\%$, $K = 5\%$, forward curve is flat equal to 5% and $n_i = 25$, $i = 1, \dots, 10$
- ✖ We have the one period caplet at time 0 $caplet_1(0)$
$$\begin{aligned} caplet_1(0) &= A \times \delta \times P(0, 2) \times E[\max(f(1; 1, 2) - K, 0)] \\ &= 1 \cdot 1 \cdot P(0, 2) \cdot 0.0020112666 \\ &= 0.90702948 \cdot 0.0020112666 \\ &= 0.0018242781 \end{aligned}$$

TABLE 1 VOLATILITY IS 10% AND STAGE FOR EVERY PERIOD IS 25

Maturity	Black	Lattice	Difference	Relative Difference (%)
1	0.0018085085	0.0018242781	0.0000157696	0.8719669656
2	0.0024348117	0.0024407546	0.0000059429	0.2440786919
3	0.0028388399	0.0028374958	-0.0000013441	0.0473484800
4	0.0031206153	0.0031282191	0.0000076038	0.2436631792
5	0.0033214311	0.0033204098	-0.0000010214	0.0307505629
6	0.0034637453	0.0034664184	0.0000026731	0.0771737036
7	0.0035616356	0.0035658574	0.0000042219	0.1185369137
8	0.0036247299	0.0036200240	-0.0000047059	0.1298286011
9	0.0036600091	0.0036633313	0.0000033221	0.0907678678
10	0.0036727489	0.0036743568	0.0000016079	0.0437804213
RMSE			0.0000063671	
1. Caplet assume $\delta = 1$ and stage 25				
2. Assume volatility is 10%, the forward curve is flat 5%				

NUMERICAL RESULTS

- ✗ Besides the relative difference, we use the RMSE to see the difference between the lattice value and Black's model for the whole maturity
- ✗ Table 2 is the results for different maturity caplets with stage 50 for each period
- ✗ We find that when we increase the stage between periods, the difference between lattice method and Black's model is close

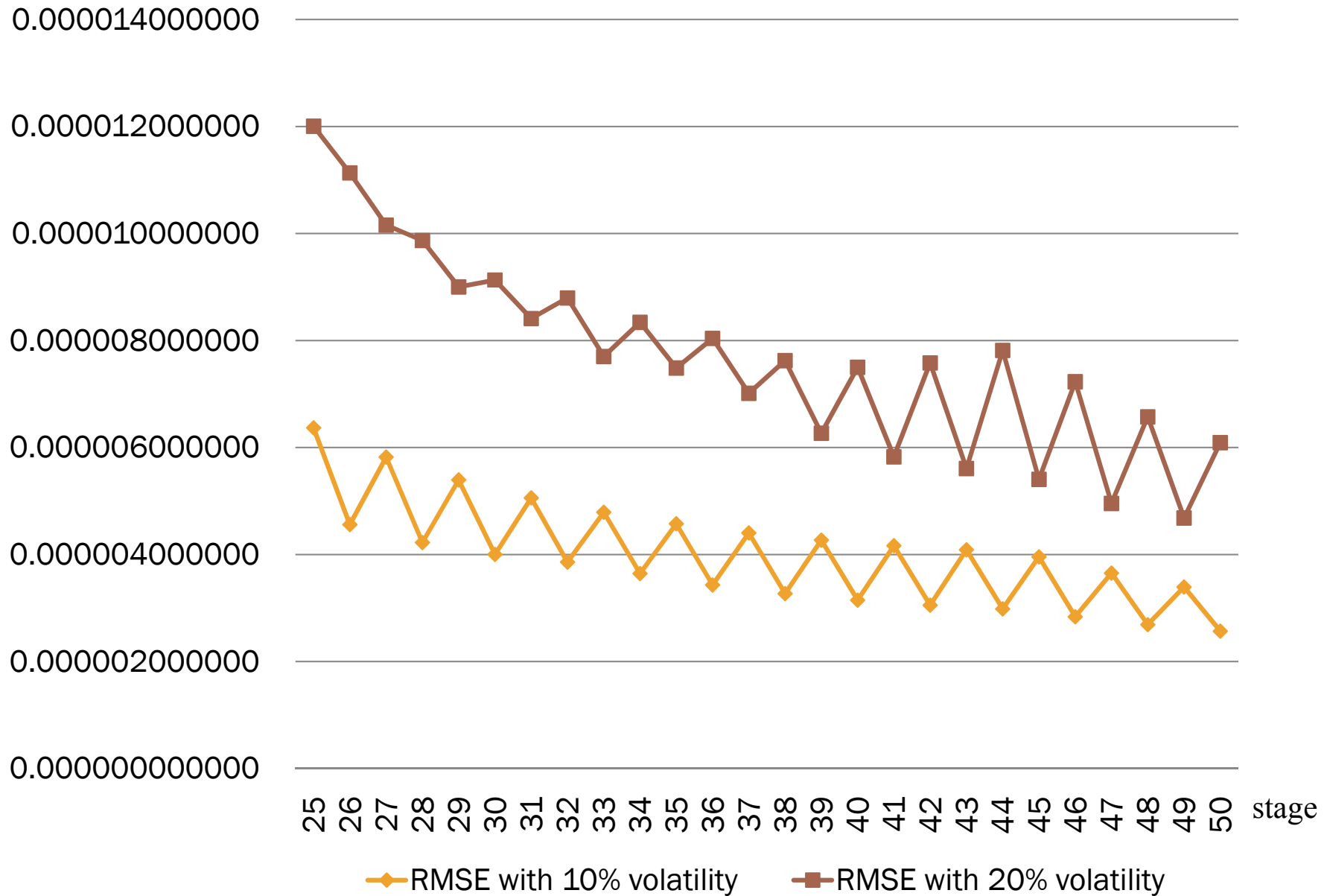
TABLE 3 VOLATILITY IS 10% AND STAGE
FOR EVERY PERIOD IS 50

Maturity	Black	Lattice	Difference	Relative Difference (%)
1	0.0018085085	0.0018099405	0.0000014320	0.0791823392
2	0.0024348117	0.0024397802	0.0000049685	0.2040608652
3	0.0028388399	0.0028434461	0.0000046061	0.1622542164
4	0.0031206153	0.0031230795	0.0000024643	0.0789673074
5	0.0033214311	0.0033207434	-0.0000006878	0.0207066243
6	0.0034637453	0.0034620815	-0.0000016638	0.0480341136
7	0.0035616356	0.0035626664	0.0000010308	0.0289416315
8	0.0036247299	0.0036267433	0.0000020134	0.0555455781
9	0.0036600091	0.0036617883	0.0000017792	0.0486107386
10	0.0036727489	0.0036734197	0.0000006708	0.0182649876
RMSE			0.0000025690	
1. Caplet assume $\delta = 1$ and stage 50				
2. Assume volatility is 10%, the forward curve is flat 5%				

CONVERGENCE BEHAVIOR VS. VOLATILITY

- ✘ We change the volatility from 10% to 20% to see the convergence behavior of RMSE at different volatility level
- ✘ We find that when the volatility at low level, the convergent rate is fast to zero. However, the volatility at high level the convergence behavior is slow

RMSE vs. Volatility



CONCLUSIONS

- ✖ Constructing the recombining binomial tree, the payoff of the interest rate derivatives on each node can be obtained
- ✖ Comparing to the theoretical value, we find the theoretical value and lattice method is close when we increase the stages.
- ✖ The sensitive analysis results of embedded option are consistent with inference of Greek letters

FUTURE WORKS

- ✖ Deriving the joint probability between the states of different binomial forward rate trees.
- ✖ Adjusting the stages between period by period to fit the strike price to reduce the nonlinearity error
- ✖ Trying to change the constant volatility to stochastic volatility to fit the volatility term structure.

The End

THANK YOU FOR YOUR LISTENING